

Durability of Consumer Products

Preventing beer bottle scuffing through surface engineering

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Abstract

Reusable brown beer bottles often reach end-of-life way before the theoretical limit of reuse cycles due to scuffing. Although the Bruin Nederlands Retourfles (BNR) has coatings in place to prevent the microscopic cracking of glass, industrial washing processes wear them off quickly. In this study, various types of new possible solutions are proposed; surface finishing techniques, coatings, surface textures and bio-inspired solutions. A test plan is described to validate the effectiveness of these solutions. Because scuffing is driven by multiple factors, the analysis concludes that a single solution is unlikely to be sufficient. A combination of techniques will be necessary to significantly extend the BNR's lifespan.

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1 Introduction

The Bruin Nederlands Retourfles (BNR) is the standard glass beer bottle in the Netherlands, allowing breweries to outsource bottle production and maintenance and streamlining the deposit process for customers. Nederlandse Brouwers, the association behind the BNR, advertises on their main page how their bottles can be reused up to forty times before the quality deteriorates below company standards, although other pages on their own site vary between twenty-five and forty. Quality standards are specified for damage, failure, and technical or visual issues [22].

Soda-lime silicate (SLS) glass is the standard material for container production due to its chemical durability, low cost, and recyclability. It is made from silica sand (SiO_2) as the main building brick, soda ash (Na_2CO_3) as a flux to reduce the melting temperature, and limestone (CaCO_3) as a stabilizer to improve chemical resistance [27].

For BNR bottles, metal oxides are added to achieve specific coloration. Green glass is obtained by adding chromium oxide (Cr_2O_3), while amber glass is produced through a carbon-sulfur process. Amber glass provides strong UV protection, blocking more than 90% of harmful light and preventing degradation of light-sensitive beverages such as beer[27].

Within these glass structures, most quality issues stem from various forms of mechanical abrasion. These include scuffing, scratching, and etching, amongst others, resulting in microscopic cracks (Griffith flaws) on the surface of the glass due to mechanical and chemical wear [3, 38]. Cracks begin predominantly at the surface of the glass, this also defines key properties such as hardness, scratch resistance, strength and chemical durability of the glass [40, 38].

Glass bottles receive polymer coatings to prevent mechanical wear; however, the industrial washing process removes part of the coating. Already in 1942 weight losses were observed as a side effect of the alkaline cleaning process and had been directly related to scuffing of bottles[20]. Back then, a patent was filed to add compounds of zinc to protect the glass against the alkali cleaning process.

However, surface coatings are not the only solution to prevent scuffing. Surface finishes that temporarily reduce friction and chemical interaction based on surface geometry pose solutions too. Therefore multiple strategies to mitigate scuffing are discussed and the following question arises:

How can surface engineering be applied to reduce scuffing on beer bottles (BNR) to increase the number of reuse cycles?

2 Research Questions

The main research question is:

"How can surface engineering be applied to reduce scuffing on beer bottles (BNR) to increase the number of reuse cycles?"

To answer this question, the following subquestions have been formulated:

1. *What is scuffing?*
2. *What factors contribute to scuffing of the BNR?*
 - (a) *How does washing affect scuffing?*
 - (b) *How does mechanical wear influence scuffing?*
3. *How can surface engineering be applied to reduce scuffing of the BNR?*
 - (a) *How can surface finishes be used to reduce scuffing?*
 - (b) *How can coatings be used to reduce scuffing?*
 - (c) *How can surface textures be used to reduce scuffing?*
 - (d) *How can biomimicry be applied?*

3 Methodology

This research is divided into three main parts. First, existing literature is analyzed to gain an understanding of the problem. This includes the mechanisms of scuffing on Bruin Nederlands Retourfles (BNR) bottles and the effects of industrial washing processes.

Next, various surface engineering technologies will be explored and proposed as possible solutions to the defined problem. The various solutions are divided into coatings, surface finishes, surface textures, and the application of biomimicry.

Finally, a test plan is proposed. This test plan is designed to test the effectiveness and feasibility of the proposed surface technology solutions.

4 Problem Definition

For a better understanding of the underlying issues, the first two sub research questions are worked out into the problem definition. First, the term scuffing is introduced; then, the two key factors to scuffing, namely mechanical wear and industrial washing, are described.

4.1 What is scuffing?

Scuffing is more than only scratching of the glass surface, it is a complex degradation process that happens at a microscopic level [35]. Even when completely new, the surface of a glass bottle is never perfectly smooth. At microscopic level, the surface of the glass bottle consists of a microscopic landscape of tiny peaks, holes, and micro-scratches [35, 11]. During the bottle's lifespan, a combination of mechanical friction between bottles and chemical etching from the washing process forces these existing microscopic scratches to grow [13, 11]. Eventually these micro scratches break through the protective coatings of the glass and begin degradation of the glass underneath [35]. As a result of this process the surface of the glass bottle becomes structurally weak, with a loose, and flaky surface layer [35]. When light hits this damaged bottle surface, it scatters and bounces in all directions. This effect creates optical interference, and is perceived as a white opaque surface, which is known as scuffing. [35, 38].

4.2 How does washing affect scuffing?

The Sinner's Circle describes four variables that interact for effective cleaning; temperature, time, mechanics, and chemistry [31]. These can all be found in the industrial washing process for glass bottles. In this process, glass bottles first go through a prewash, a low-temperature wash that removes large amounts of pollutants. Afterwards, the bottles go through various caustic baths at a variety of temperatures, after which these chemicals are rinsed off [11].

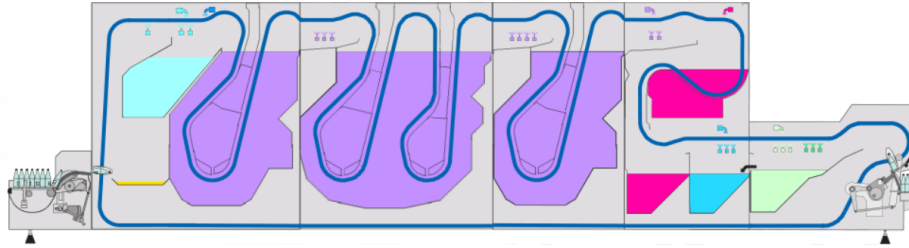
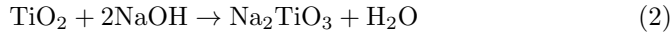
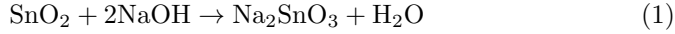


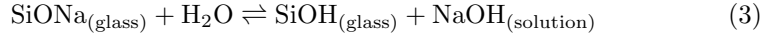
Figure 1: Industrial washing program [11]

The industrial washing program relies on Caustic Soda (NaOH) for effective cleaning [21], however, this solution also reacts with glass and its coatings. Two possible chemical reactions can occur when Caustic Soda comes in contact with glass. But before these reactions with glass occur, the first washing cycles can also aid in the wear of the films around the glass.

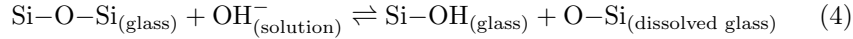
There are two main coating layers: a hot-end coating increasing mechanical strength and adherence of the lubricating cold-end coating. The outer cold-end coating, typically consisting of organic materials such as soaps, waxes, silicones, or polyethylenes, quickly deteriorates during the first caustic baths. The inner hot-end coating typically consists of oxides such as SnO_2 or TiO_2 , these slowly deteriorate over repeated caustic baths. TiO_2 will require high temperatures to react, which is present in some of the caustic baths for proper cleaning. Some possible reactions include [35];



During the initial phase of glass corrosion, static aqueous corrosion occurs through surface moisture entrapment, and dynamic aqueous corrosion is created by condensation run-off, restarting the reaction below [11].



The bonding between the silicon and oxide atoms weakens in the second phase of corrosion, dissolving the glass.



4.3 How does mechanical wear influence scuffing?

Throughout their lifespan, returnable glass bottles experience constant mechanical wear, which is the biggest cause of scuffing. This damage is shown as whitish marks or rings where the glass has been degraded both physically and chemically [11]. Inside washing and filling facilities, bottles scrape against one another and against the machinery. The friction is most severe during single-line inspections and when bottles are fed into or discharged from the equipment [2]. As a result of this handling, wear and tear concentrates exactly where the bottles make the most contact, namely at the shoulder (top), and the base (bottom) [13], as shown in Figure 2. Note that the middle of the bottle is also noted in the figure, however, this is not a primary scuffing location.

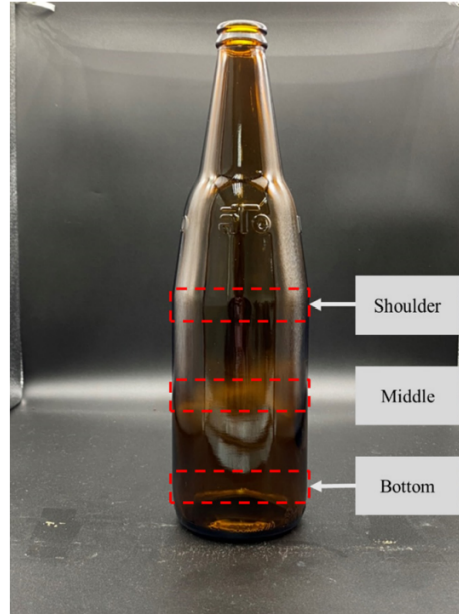


Figure 2: Locations of bottle contact points [13].

As the number of washing cycles increases throughout the bottle's lifespan, the impact of mechanical wear also increases. High temperatures and strong alkaline detergents gradually scrape away the bottle's coatings, whilst at the same time, breaking down the chemical structure of the glass itself (section 4.2)[11]. Once these protective layers are eroded, the bare glass beneath is subjected to the impacts caused by friction. This direct friction leads to the formation of micro-cracks, which are surface flaws smaller than $100\text{ }\mu\text{m}$ [13]. Because glass is a brittle material, these micro-cracks act as critical stress concentrators that significantly reduce the bottle's overall strength and its resistance to internal pressure [13, 38]. Furthermore, mechanical shocks caused during transport in combination with the internal pressure of refilling carbonated beverages, causes these initial micro-cracks to expand. This leads to a degradation of the structural integrity of the bottle. This continues until it is prone to failure, or the white scuffing marks cause the bottle to no longer meet visual quality standards [11, 13].

5 Proposed Solutions

Now that the concept of scuffing has been defined, and the key influences that lead to scuffing are described, a deep dive is performed into possible solutions. Four types of solutions are worked out here; surface finishes, coatings, textures and solutions found from examples in nature.

5.1 How can surface finishes be used to reduce scuffing?

While coatings act as an external layer to reduce friction, surface finishes alter the soda-lime-silica glass substrate itself. During the washing cycle and on the bottling line, bottles face high contact forces as described in 4.3. Besides this, the protective coatings are also washed off, this results in the bare glass experiencing much more friction. This high friction creates microscopic cracks [13], and lots of concentrated cracks cause scuffing. These cracks however, can also act as Griffith flaws. These are critical stress concentrators within the glass surface. The flaws cause the glass to fail at stress levels far below its theoretical strength [9].

To prevent micro-cracks from propagating on the glass surface, either the friction can be lowered or the glass must be hardened. Surface finishes are unable to reduce friction, but are able to harden the glass surface. They alter the physical properties of the surface itself to prevent micro-cracks from expanding and causing scuffing of the glass [16].

5.1.1 Topological Surface Modifications

One way of preventing micro-cracks and Griffith flaws from growing on the glass substrate is to melt or dissolve them away. This can be done using thermal and chemical smoothing techniques. These flatten the surface of the glass and improve the contact resistance of the surface.

These techniques remove asperities and pre-existing Griffith flaws which suppresses the formation of new micro-cracks. This is achieved because mechanical impacts are spread out over a larger flat surface area rather than being concentrated at pre-existing Griffith flaws. The glass can therefore withstand higher forces without failing [16].

Thermal smoothing, such as high-intensity plasma-arc heating or more traditional fire polishing, rapidly heats the outer surface of the bottle to its softening point. This allows the glass to locally flow and fuse resulting in a flat topology [37]. High-intensity plasma-arc heating is also an effective possibility for industrial use as it is capable of processing large amounts of glass quickly, however, it requires high amounts of energy [37].

Chemical smoothing makes use of corrosive agents to dissolve away the cracked outer layer of the glass. By immersing the glass in solutions of etching mixtures, the top layer of the glass is dissolved. This way, the Griffith flaws previously present on this surface are dissolved too. This exposes a fresh surface of flat glass beneath [34]. While this process effectively flattens the surface to remove cracks, it is often difficult to implement this process within a bottling environment due to the hazardous nature of the reagents.

5.1.2 Surface Modification Via Hydrothermal Treatment

Another technique to strengthen soda-lime-silica glass is treating it with superheated steam. The main advantage is that it modifies the structure without requiring the removal of any material or melting the surface. When glass is treated with steam at 510°C (a temperature just below the point where the glass starts to soften), the hardness just below the surface increases by about 30%. This change only happens in an extremely thin top layer of up to 300nm deep [28]. It should be noted that this data comes from an unpublished source, and these findings have not yet been independently verified. The results should therefore be interpreted with caution until they are verified.

The increase in hardness is due to a physical process that alters the internal structure of the glass. The steam causes the bonds in the glass network to break down slightly; this leads to network depolymerization. In this stage the silica network becomes looser, microscopic channels are created in the top surface of the glass. Water molecules from the steam use these channels to enter the glass and fill small empty spaces known as interstitial voids. These water molecules secure themselves in the voids using hydrogen bonds. Because the empty spaces are now filled and supported by the water, the surface becomes harder [28, 39].

The downside of this technique is that the added water does not naturally stay inside the glass when exposed to ambient air. Over time, as the glass ages, the water molecules escape back into the outside atmosphere. As soon as the water leaves these small spaces, the extra subsurface hardness of the glass decreases again. To prevent this water from escaping into the atmosphere, a coating can be applied directly to the glass surface after the steam treatment [28].

5.1.3 Surface Modification Via Ion-Exchange

Ion-exchange (IOX) is a process that swaps smaller ions for larger ones to induce a compressive stress in the surface of glass. It works by submerging the glass substrate in a molten salt bath, usually potassium nitrate (KNO_3). The temperature of this bath is just below the glass transition point. During submersion, the surface is strengthened through diffusion. The smaller sodium ions (Na^+) from the glass are replaced by larger potassium ions (K^+) from the salt bath [10]. Because the larger potassium ions are forced into the spots previously occupied by the smaller sodium ions, they create a compressive stress at the surface. This compressive layer within the surface acts as a physical barrier. This

barrier prevents Griffith flaws to grow, as the applied mechanical stress must first overcome this internal compression at the surface. Furthermore, the KNO_3 can also reduce existing cracks, causing the glass on either side to expand and effectively 'repair' the flaw by squeezing it shut [14].

5.2 How can coatings be used to reduce scuffing?

In order to prevent scuffing, glass should resist plastic deformation as much as possible, aiming for a high strength. However, a strong material can be brittle and prone to shatter. To prevent shattering, a release of external stresses is required through toughness [26].

Currently, to maintain both strength and toughness, glass bottles contain two coatings. The first one is a hot-end coating to increase mechanical strength and adherence of the cold-end coating, which reduces friction and increases toughness. In 4.2 the reactions between these coatings and the caustic baths are described.

Can an extra coating or a change in coating prevent the reaction between the caustic bath and the coatings, or further prevent mechanical wear? Keeping in mind that the coating has to be food-safe, and will not reduce other mechanical aspects of glass, such as lubrication or heat resistance.

5.2.1 MXDA-Organosilicon-Epoxy-Resin (MXDA-OSER)

A triethoxysilane (GPTES) coating combined with the curing agent m-Xylylenediamine (MXDA) has multiple characteristics relevant to bottle scuffing. Firstly, the optically highly transparent coating has a double cross-linked structure, making it quite hard and scratch resistant, with a Young's modulus of around 10.8 GPa [25].

Secondly, with the addition of a hexamethyldisilazane (HMDS) soak, the coating significantly reduced surface energy to a contact angle of 116° [25]. This hydrophobicity could prevent the entrapment of droplets from the caustic bath.

The chemical stability of the coating in alkaline environments was found to decrease the cross-link structure. However, this was tested at a pH of 12 over the timespan of a week, where no impact was measured after a day on a pencil hardness test [25]. The industrial washing process for returnable bottles takes around 15 minutes at a pH level above 9.[11] No exact calculations can be made, given the data. However, a crude guesstimation of 24 hours divided by 15 minutes gives 96 washing cycles without any impact on hardness, more than doubling the amount of reuses a BNR can achieve [22]. It is important to note that the effects of heat and mechanical interaction are not tested simultaneously with the alkaline environment, a combination of factors could greatly influence results.

Finally, the coating can withstand a maximum 155°C for 12.7 minutes, which is way above the maximum of 78°C of the industrial washing process [11].

An MXDA-OSER coating has the hardness, hydrophobicity, chemical stability and temperature resistance to prevent scuffing, and could be introduced as an alternative for the hot-end coating currently applied on returnable bottles. As toughness will still be required, the outer cold-end coating should still be maintained.

The surface of the MXDA-OSER should allow adherence to the cold-end coating. The coating MXDA-OSER was found to be very smooth, with a roughness of 0.274 nm root-mean-square deviation, limiting mechanical interlocking. A coupling agent could be required to promote adhesion between the coatings [30].

Some of the contents for preparing an MXDA-OSER coating are harmful when touched, swallowed or inhaled[33]. MXDA-OSER is marketed for commercial applications[25], but its food-safety has not yet been tested.

5.3 How can surface textures be used to reduce scuffing?

Surface texturing is about changing the surface of soda-lime silica glass at the micro- and nanoscale. This is done to improve its frictional and chemical behavior. For the BNR bottle, scuffing is not caused by one single contact event. But, it develops over time through a combined mechanical and chemical process.

By adding specific surface patterns into the glass, such as micro-dimples or grooves, it is possible to influence how the bottle reacts to both mechanical and chemical wear in industrial cleaning. These textures can reduce the actual area of contact, trap abrasive particles, and change how liquids interact with the surface.

5.3.1 Contact Area Reduction and Stress Distribution

According to basic adhesion theory, the friction force during dry sliding is related to how much of the two surfaces are actually in contact with each other. After production, the surface of glass is never perfectly flat. It contains small surface deviations: both negative deviations, such as micro-cracks and positive deviations, such as asperities. When the glass slides against another surface, these surface deviations can grab onto each other and resist motion [1]. This resistance to motion causes local forces in these asperities to become too high, causing surface deviations to grow. And to the formation of micro-cracks or Griffith flaws, that become visible as scuffing.

Introducing a controlled surface texture, like patterns made of micro-dimples, changes how contact between glass surfaces works [36]. By creating these dimples, a portion of the smooth surface is replaced by empty space. When the bottle slides against an object, the solid-to-solid contact area is reduced. As a result, the friction force is also reduced [36].

In addition, chemically etched surface textures help distribute the stress at the contact points. Because material is removed as dimples are created, the effective macroscopic contact area is reduced. This initially leads to higher local contact pressures on the remaining asperities. However, the micro-features induce localized elastic deformation, making the contacting surface more mechanically forgiving [17]. This increased local deformation allows the contact stresses to be distributed much more evenly across the surface. This reduces the peak stress concentrations, and minimizes the chance of fracture [17].

5.3.2 Debris Entrapment

During bottle contact (with other bottles or machinery), small fragments of glass, or other materials, can break off from the surface as a result of friction and local stress peaks [29]. These fragments, create a three-body abrasive wear system between the bottle, the fragments, and another bottle or object. When these fragments remain between two sliding surfaces, they can act as abrasive particles. Instead of smooth sliding, the debris scratches the glass surface, which accelerates the formation and growth of micro-cracks and scuffing[29, 15].

On a smooth, untextured glass surface, there is no place for this debris to escape. The particles remain trapped in the contact area and continue to damage the surface. This creates a self-reinforcing wear mechanism, in which each contact produces more debris and thus more damage to the bottle [5].

Surface texturing can reduce this effect by providing spaces where debris can be captured. Micro-dimples and grooves act as small collectors that trap the loose particles and prevent them from staying in the active contact area [29]. When a textured bottle slides against an object or other bottle, abrasive debris is more likely to be captured into these textured parts instead of being dragged across the surface. As a result, the abrasive interaction between the two glass surfaces is reduced.

5.3.3 Fluid Film Lubrication

Bottling line conditions are never fully dry; in industrial washing plants and during production and transport, bottles are exposed to high humidity, condensation, residual washing water, and conveyor lubricants (section 4.2). As a result, bottle-to-bottle and bottle-to-conveyor contact frequently occurs in the presence of lubrication. Under these conditions, surface texturing can provide an additional mechanism for reducing scuffing by influencing the behavior of this fluid film between the objects.

Micro-dimples and grooves on the glass surface can reduce scuffing when a thin liquid layer is present. During sliding, this liquid flows over the textured surface and is drawn into the small dimples and grooves [42]. When the liquid flows over a micro-dimple, it has to follow the shape of the surface. As it leaves the dimple, the gap between the surfaces becomes smaller again. This narrowing slightly squeezes the liquid, which causes the pressure to rise locally. This increase in pressure is known as the hydrodynamic wedge effect [42, 19].

When many of these micro-features are distributed across the glass surface, the small pressure increase generated at each dimple combine. Together, they can produce a net hydrodynamic lift force that slightly separates the two surfaces. This increases the thickness of the fluid film, reduces bottle-to-object contact, and lowers friction thus the risk of scuffing [42, 19].

However, there are also downsides to adding micro features to the BNR. While it is beneficial to reduce friction, it should be noted that by keeping prolonged liquid retention on the bottle’s surface may also introduce negative chemical effects. In returnable bottle systems, industrial washing processes rely on strongly alkaline cleaning solutions. These solutions can accelerate chemical attack, leading to surface etching and the growth of existing micro-defects (section 4.2). If surface textures retain cleaning liquids for longer periods, local high concentrations of aggressive chemicals may occur [4]. This further promotes glass degradation and scuffing. Therefore, the effectiveness of surface texturing is based on a balance between lubrication, whilst avoiding long-term retention of cleaning agents.

5.3.4 Surface Wettability Control

Surface texturing can also be used to protect the textured glass from aggressive washing agents. This can be done by changing the surface wettability from hydrophilic to hydrophobic. A hydrophilic glass surface naturally attracts water, causing cleaning fluids to spread and remain on the surface for extended periods. A hydrophobic surface on the other hand, limits liquid adhesion on the glass surface and promotes liquids to run off [32].

The wettability of a textured surface depends on the surface energy of the material. By creating a specific and careful dense pattern of micro-dimples, the glass surface can be changed into a non-wetting Cassie–Baxter state [32]. In this state, small pockets of air remain trapped within the dimples of the texture. As a result, liquids are unable to fully enter the surface (see Figure 3).

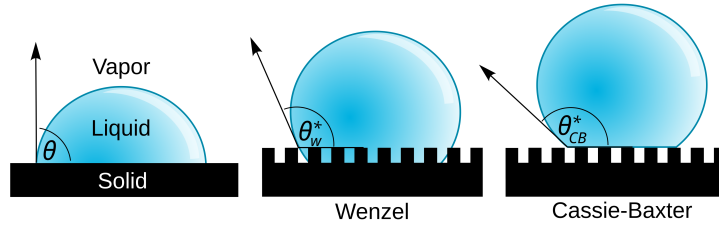


Figure 3: Schematic illustration of wetting states on glass surfaces. (A) Young wetting on a smooth, untreated surface with contact angle θ . (B) Wenzel wetting on a textured surface, where the liquid fully penetrates the surface features, resulting in a contact angle θ_W . (C) Cassie–Baxter wetting on a textured surface, where the liquid rests on the surface peaks with air trapped beneath, leading to a higher apparent contact angle θ_{CB} [41].

In the Cassie–Baxter state, the high contact angle causes the alkaline droplets to be repelled off the surface. They roll off during the rinsing and drying stages instead of remaining on the bottle. This reduces the time of cleaning agents making contact with the surface and thus minimizing the damage caused.

Experimental studies have created superhydrophobic surfaces using precise laser-textures. These studies show that these surfaces maintain their stability under alkaline conditions and protect the material against corrosion. The material was protected even after prolonged exposure to high-pH solutions [18]. This suggests that the surface texture can protect the glass substrate from chemical attacks even after multiple washing cycles.

5.4 How can biomimicry be applied?

By closely inspecting the mechanisms nature uses to tackle problems, potential solutions can be found for engineering challenges. Here, these mechanisms are grouped into hydrophobicity, mechanical resilience, and alkaline protection.

5.4.1 Superhydrophobicity

One often applied example in biomimicry is the Lotus leaf's superhydrophobicity. Through a twofold system, the Lotus leaf maintains a bacteria-free system. The first part of this system is the surface texture, containing papillae that lower the contact angle to below 160 degrees [23]. These papillae are of varying heights, further lowering adhesion [7]. Secondly, a wax layer is coated on top of the leaf, which further aids in the superhydrophobicity of the Lotus leaf [6]. Superhydrophobicity could prevent droplets from the caustic bath from sticking to the surface of the bottle, which would otherwise continue to damage the bottle's coating or even the glass underneath.[24]

5.4.2 Mechanical Resilience

The inner shell layer of the Fan Mussel (*Pinna Nobilis*), has unique mechanical properties. The shell has flexible molecular bonds, allowing it to plastically deform and absorb energy by unfolding proteins, the shell is able to refold its proteins when the external stress is removed. Even after fracture, crack propagation is prevented, as the macroscale architecture stays intact [8]. A similar system could be created by adding a few layers of a relatively strong material (the *Pinna Nobilis* shell uses CaCO_3) in pseudo-hexagonal shapes, interlocking in a brick-and-mortar style, bonded together by a slim organic layer. This system would prevent the glass bottle from crack propagation, and therefore prevent scuffing.

5.4.3 Alkaline Protection

Many organisms live in extreme alkaline environments, maintaining homeostasis in places with pH levels in a range from 10-13. Alkaliphiles are microorganisms that maintain growth at high pH levels. They have several mechanisms in order to maintain a neutral internal pH level. One mechanism is an acidic outer wall, this negatively charged wall reacts to incoming hydroxide ions, preventing them from reacting with the inner parts of the microorganism. A second defense system is a Na^+/H^+ antiporter, this membrane protein regulates pH by using membrane potential. To achieve a similar effect, the BNR could include a layer of higher acidity, however, this could have negative side effects for the user, the transport system and other parts of the bottle's lifecycle. An antiporter helps regulate the in- and output of molecules, but no such thing is required to protect the glass, as it would prefer not to allow anything through [12].

Table 1: Overview matrix of proposed solutions

Category	Solution	Mechanism	Advantages	Disadvantages
Surface Finishes	Plasma-arc heating	Melts the surface locally to heal Griffith flaws.	Fast and scalable.	Requires high amounts of energy.
	Chemical smoothing	Uses corrosive agents to dissolve the damaged outer layer and remove Griffith flaws.	Reduces surface roughness to a minimum.	Difficult to implement industrially due to hazardous reagents.
	Hydrothermal	Fills glass voids with water via superheated steam.	Glass surface becomes harder and more durable against Griffith flaws.	Reversible; requires coating to trap water.
	Ion-exchange	Molten salt bath replaces Na^+ with larger K^+ ions.	Creates a compressive barrier against crack growth and heals existing cracks.	Requires high-temperature molten salt bath.
Coatings	MXDA-OSER	Transparent double cross-linked epoxy.	Optically transparent, scratch-resistant, hydrophobic, chemically stable, temperature resistant.	Unproven food safety; requires cold-end coating adherence.
Surface Textures	Micro-dimples & Grooves	Micro/nanoscale physical alteration.	Reduces solid-to-solid contact area, distributes stress, traps debris, lubricates, repels cleaning agents.	Unproven cost, scalability, producibility, and long-term durability.
Biomimicry	Lotus leaf	Imitates natural superhydrophobicity	Repels caustic drops from continuing corrosion	Unproven cost, scalability, producibility, and long-term durability.
	Fan mussel	Imitates natural resilience.	Prevents crack propagation.	Unproven cost, scalability, producibility, and long-term durability.

6 Test Plan

As mentioned, scuffing happens as a result of mechanical wear and chemical instability. For a surface engineering solution to prove effective in increasing durability, resilience to one or both factors will have to improve.

For testing purposes, the different proposed surface engineering solutions are applied onto 5 by 5 centimeter square glass sheets with a 5 millimeter thickness. For comparison, a baseline of a glass sheet receiving treatment similar to the BNR's treatment is added, as well as a sheet of untreated glass for control. For the chemical stability test fifteen samples of all the variously treated sheets of glass are prepared, and for all the others, five samples of all the variously treated sheets of glass are prepared.

The test plan is divided into three main parts; First, the chemical stability in alkaline environments of the different treated pieces of glass is tested. Next, the scratch-resistance of the different treated pieces of glass is tested. Finally, a combination of factors is tested.

6.1 Chemical stability

Although not all proposed surface engineering solutions focus on chemical stability in alkaline environments, all are tested to compare against the baseline of the BNR's treatment.

All samples are weighed with milligram precision. Fifteen samples, divided into three groups, of all the variously treated sheets of glass are soaked in an alkaline solution with a pH of 10. The first group is removed after 24 hours, the second after 48, and the last after 72. After the samples are taken from the solution, they are rinsed and dried. Again, the weight of the samples is measured with milligram precision.

The weight change will give an indication of the glass sheets chemical stability in alkaline environments, no weight change should be observed if the treatment makes for a completely chemically stable solution. For every solution, the null hypothesis will be tested.

H0a: The surface engineering solution does not affect chemical stability of glass in alkaline environments.

6.2 Scratch resistance

To determine the scratch resistance of the various proposed solutions, a Rockwell C diamond stylus is put onto a scratch tester. A load of 10 N is applied, and the tester moves 2 centimeters over the glass sheet at a speed of 5 mm/s. The scratches are then measured using a confocal microscope.

The wear is quantified using Archard's equation; $K = \frac{\Delta V}{F_N * S}$

ΔV = Volume removed

F_N = Load (10 N)

S = Sliding distance (20 mm)

H0b: The surface engineering solution does not affect the scratch resistance of glass.

6.3 Combined test

A final test combines both test setups, to create a scenario close to the industrial washing process. First, the samples are soaked in an alkaline solution with a pH of 10 for 24 hours. Then, the samples are rinsed and dried, and directly after, the scratch test is performed once again.

H0c: The surface engineering solution does not affect the scratch resistance of glass in alkaline environments.

7 Discussion

This study is based on a literature review and concept analysis. Within this research, no new experiments were conducted. With this approach, a broad understanding of scuffing mechanisms and surface engineering techniques are explored. However, there is not enough research available to directly compare all the different proposed solutions. Therefore, the study has the the following limitations:

First limitation is the limited availability of detailed industrial data. Bottling and washing processes are often commercially sensitive, which makes it difficult to access precise process parameters and bottle specifications. Existing research mainly relies on laboratory-based studies, these studies do offer insights into scuffing behavior and surface modifications. But translating this data to real world environments for the BNR, requires further research.

Second limitation is that long-term durability of surface-textured glass is not evaluated. While surface texturing shows promising results, expected to be more durable than coating-based solutions, the effects of repeated mechanical contact and prolonged exposure to alkaline cleaning solutions on the texture itself remain unexplored.

The third limitation is lack of discussion about the combination of different solutions. The proposed surface finish techniques, coatings and surface textures are independently researched through literature. Researching the a combination of multiple techniques is something that should be further investigated in future research and was out of the scope for this research.

A last limitation of this study is the lack of thorough analysis of industrial feasibility of the proposed solutions. Factors such as production costs, (large-scale) manufacturability, and environmental impacts were not assessed. This is mainly due to most research is still in laboratory phase and not applied to real world applications yet that align with the BNR.

7.1 Future Research

Future research should focus on experimental testing of surface-textured BNR bottles under realistic washing, transport, and reuse conditions. Long-term durability testing over multiple reuse cycles would define whether surface textures remain effective, or whether repeated mechanical and chemical exposure gradually degrades the features.

An open question that follows from this is what the optimal dimple geometry actually looks like for this specific application. Based on the literature, depth-to-diameter ratio, dimple density, and edge geometry all influence the balance between contact area reduction, debris entrapment, and hydrodynamic lift. Since these mechanisms require different dimple geometries, the ideal shape likely requires a design that combines multiple characteristics. What this looks like in practice for soda-lime silica glass under alkaline washing conditions is something that would need dedicated testing to answer.

Beyond texturing, future work should also investigate combinations of solutions. In this report, surface finishes, coatings, and textures were each analyzed independently. A combined approach, for example, ion-exchange strengthening of the glass substrate together with a textured hydrophobic surface, may perform better than a single method on its own. Whether such combinations are manufacturable and economically viable at BNR scale remains an open question for further research.

8 Conclusion

Scuffing is the result of mechanical and chemical wear acting on the glass surface over repeated reuse cycles. Each time a BNR bottle goes through the washing process, alkaline cleaning agents strip away its protective coatings and gradually attack the glass surface underneath. With the coatings gone, mechanical friction has direct access to the bare glass, allowing micro-cracks to grow and propagate. Wear debris caught between sliding surfaces makes this worse, and eventually the bottle either breaks or the white scuff marks push it below visual quality standards.

Several surface engineering strategies can slow this process down. Surface finishes like ion-exchange and hydrothermal treatment strengthen the glass itself, making it harder for cracks to propagate. The MXDA-OSER coating may offer chemical resistance and hydrophobicity. However, it is not yet verified on food safety and this is an important factor. Surface texturing stands out as the most promising option, it targets multiple scuffing mechanisms at once: reducing contact area, trapping debris, and controlling how liquids interact with the surface. But, industrial feasibility should be further researched in order to be able to apply this to the BNR. Finally, biomimicry offers useful inspiration, particularly the Lotus leaf and Fan Mussel, but direct applications to the BNR remain limited.

None of these solutions has been tested for real BNR conditions. It is unlikely that one measure alone will be enough to prevent scuffing. The process is caused by too many factors at once for that. A combination of techniques is likely to be necessary, and the test plan proposed in this report provides a logical first step to explore them. These results may propose future research directions to get closer to a more durable BNR bottle.

9 AI Disclosure

During this research, several digital tools were used to support and assist in literature exploration and improve clarity of writing. LLM tools such as Microsoft Copilot, Gemini, and ChatGPT were used to help find relevant publications and articles, and to explore ideas and refine wording. Google Scholar Labs and Consensus were used to help in the search of academic literature. At last, Grammarly was used for grammar and spelling checks.

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